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Walking in Every Direction: Aperiodic Training Data and Anisotropic Kernels for Step Movement Analysis

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Abstract

Human movement is not merely present or absent, nor simply fast or slow. At its core, it has a direction, and this direction carries information that most machine learning (ML) models trained on movement data currently ignore. Convolutional neural networks (CNNs) built from regular, square kernels are good at recognizing the content of a photograph or an audio sample, where the patterns to be detected are not inherently directional. Human movement is different: following a path is a directed action and, thus, unlikely to produce a symmetric pattern. Consequently, a CNN with a non-regular kernel can be trained to capture something closer to intention than mere presence. We apply this idea to sensor networks that record such movements without cameras: floor sensor mats that pick up unidentified footsteps in public spaces such as buses or supermarkets. Instead of partitioning the floor section with sensors like a chessboard, we enrich our training data using a floor layout derived from an aperiodic tiling. This way, we produce synthetic movement data whose walking directions are free of an angular bias. We combine it with a detection network built from direction-sensitive, non-square convolution kernels. On real recordings, synthetic enrichment clearly improves footstep detection over training on scarce real data alone. For two feasibility case studies, one in a grocery store and one on a city bus, we show how analyzing the direction of people's footsteps simplifies AI-supported planning, while relying only on the anonymos foot movement data from a narrow floor section covered by a sensor mat.

Keywords

gait analysis, aperiodic data partitioning, anisotropic convolution, situated intelligence

1 Introduction

In this article, we present techniques for teaching an AI model to understand movement data using the example of footstep recordings from sensor mats. The AI model should not primarily find the destination of people stepping over the sensor mats but rather come to an interpretation of the recorded directed moves. Interpreting step moves means distinguishing a footstep from noise, a hurried walk from a leisurely one, and a queue from a passing crowd. The direction actually taken is not arbitrary: it is exactly the information we need. Any path takes a walking person somewhere different, and, and, as we argue throughout this work, teaching an AI model to notice this difference, rather than treating every direction alike, is a promising way to make movement data more informative.

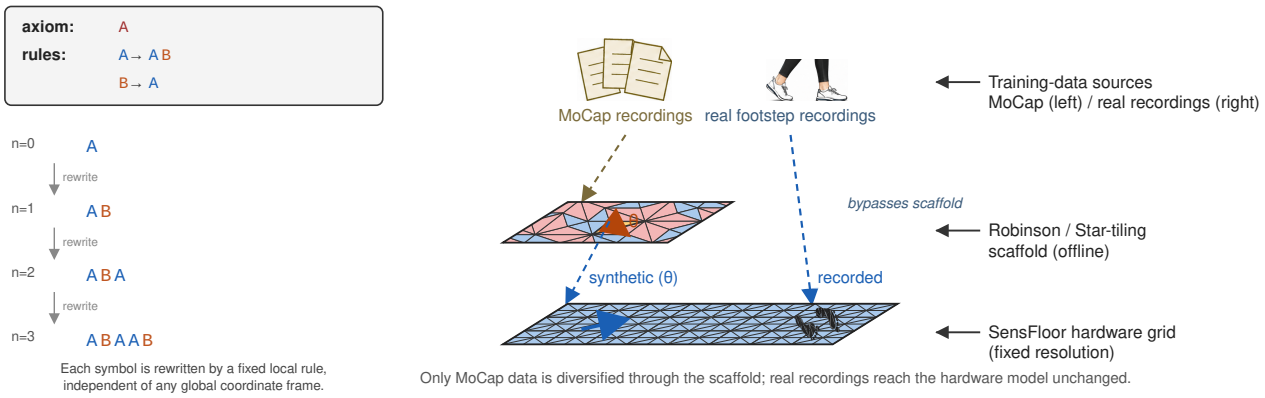
Before we come back to our methodology for movement data analysis and the importance of direction-awareness, let us motivate our two case studies: Most people will agree that it is a good idea to align bus schedules and the placement of new bus stops with the actual needs of passengers.

However, for city councils, it is not trivial to determine where and when travelers would like to have a new stop. Therefore, the planning of new routes is typically based on map partitioning and demographic data[1]. Crowding around the bus entrance or other dynamic data, such as average travel times, is generally ignored since this information is not acquired when only ticket sales data are collected. Therefore, no passenger behavior analysis is conducted, although it could substantially improve the planning of bus routes[2].

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(a) L-system rewriting **(b)** Two training-data sources converging on the hardware grid

Figure 1. Synthetic movement data that look real: the principle goes back to (a) L-systems[3] that rewrite a seed via a fixed rule. Robinson[4] and Penrose[5] applied this idea geometrically to build aperiodic star tilings. (b) We use two data sources: MoCap recordings (top left) pass through the star-tiling scaffold (middle) diversifying it to θ, while labeled recordings (top right) bypass it.

Grocery store management similarly depends on understanding customers: Foreseeing what they buy next helps avoid food waste, and thus, sustainable inventory management. Sales and visitor data, broken down by department, can already train AI models to make such predictions[6].

Despite these benefits for urban mobility and sustainability, most people will not agree to being filmed on a bus or in a store to create AI training data. Recording and processing video of individuals is often refused or even prohibited by law[7]. Tracking people in public places like buses is therefore typically done via on-site surveys[2], which suffer from coverage errors: relevant groups, such as people in a rush or dissatisfied customers, often do not participate[8].

Our case studies show that an AI model trained to detect and characterize local movement patterns does not need to identify individuals and is a suitable alternative to filming or questionnaires. The gait data is obtained via floor sensors collecting the footstep data any person leaves. These data are not privacy-sensitive, can be recorded in public places without consent, and still yield verifiable, meaningful behavioral information, as proven in our experiments.

The training data we use are partially synthetic, but realistic as the step-data generation is based on MoCap (motion capture) recordings. MoCap recordings are human movement data, publicly available for everyday actions such as walking and running[9]. Given a starting position, an orientation vector is required for replaying a MoCap recording that drives a biomechanical simulation imitating a pedestrian walking towards the respective direction. In our initial experiments on generating synthetic step data by simulating humans walking over our sensor floor, we simply set one heading per pedestrian, pointing inward from the entry side, with a random offset of $\pm 60°$ [10]. Enriching real step data with such imitated footsteps made the detection model significantly more accurate than a previous version trained solely with real data. Another random, but equally distributed, choice of orientation vectors is polar sector selection: the $360°$ around the center point of the current rectangular tile is divided into n equal angular sectors, each labeled by a compass direction. With $n = 6$, this yields two $60°$-open triangles (north and south) and four $60°$-open trapezoids (north east, south east, south west and north west). For computing the next step in the MoCap simulation of a person walking from one tile to the next, one of the 6 possible bisectors of the $60°$ angles in the star shape resulting from partition of the tile the person stands on is picked randomly. These are exactly the six directions from one cell of a honeycomb to its neighbors. However, both orientation selections are biased: humans do not walk like bees on a honeycomb.

As an extension, in this article, we diversify the MoCap data by projecting them through a scaffold with a Robinson tiling, as shown in Fig. 1, to set the orientation of the simulated steps. The directions of this tiling's triangles lead to a choice of walking directions for the synthetic data generation that is free of the angular bias of our earlier sampler.

For a direction-aware step detection we also use CNNs with non-square convolution kernels. That combination of direction-aware data enrichment and classification is intended to get us closer to understanding the differences in how people actually walk than a purely data-driven step analysis that ignores the intentions inherently contained in step recordings. Summing this up in two research questions, we study a) if floor sensor mats can be used to reliably detect and attribute footsteps to individuals; and b) does direction-unbiased synthetic training data improve this.

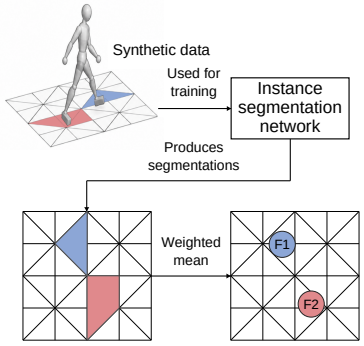


Figure 2. The neural network is trained on synthetic data and produces instance masks. Masks are converted to feet positions using a weighted mean.

We address our research questions using a tracking pipeline of instance segmentation (see Fig. 2) and direction-sensitive kernels (Subsection 3.4), trained on simulation data enriched with our Robinson-triangle scaffold (Subsection 3.2). Our experiments answer a) positively; for b), they show a clear benefit of synthetic enrichment as such, but the expected benefit of direction-unbiased enrichment combined with direction-sensitive kernels was not measurable on our limited real test set (Subsection 4.3).

The methodological considerations presented in this article are underpinned by real-world experiments: We installed floor sensor mats in a grocery store, near a digital billboard, and in a city bus, where timing and movement data showed clearly when the observed places were empty or overcrowded. To prepare the experiments, we recorded ourselves walking across the sensor mats in each environment; combined with simulation data, this let us train a reliable model. We report instance-level recognition metrics and show how indicators like length of stay and peaks are derived.

We cover related work in Section 2. Our network and its synthetic-data pipeline are described in Section 3. Two methodological choices set this pipeline apart: an aperiodic, direction-unbiased training-data scaffold built from Robinson triangles[11] (Subsection 3.2), and our direction-sensitive convolution kernels (Subsection 3.4), the latter trustworthy only when trained on the former's unbiased data. The experimental evaluation and conclusion follow in Section 4.

2 Related Work

One branch of data science focused on behavioral data[12] is *process mining*. One problem of ERP software, as it is often used in process mining for business, is that the captured data is too coarse-grained for the detection of behavioral patterns. One reason for gaps in the data is that many systems retrieve data using *pull*-methods importing measurements on demand. Our floor sensors produce small data packets, which can all be *pushed* to our backend, leading to a more reliable measurement interpretation[13].

Alternatives to floor sensors that are sometimes used for monitoring warehouse employees are shelf sensors and wearable beacon tags, worn on the wrist, chest or helmet[14]. While process mining can benefit from the combination of multiple data sources, maintaining situational awareness becomes more difficult with the number of systems involved[15]. Other approaches to action tracking use WiFi beacons[16] or Bluetooth Low Energy (BLE)[17]. Similar to the warehouse monitoring scenario, all of these setups require the tracked persons to carry some kind of transponder device: a smartphone, a beacon tag, or a badge which is workable on a construction site or in a supervised care setting[17], but not in the anonymous public spaces we target.

An AI model for anonymous foot localization and local tracking, as presented in this paper, was not used in the studies discussed above, some of which also combined sensor and camera recordings. We only use floor sensors and our AI model ignores identities, i.e., we focus on anonymous local movement patterns instead of recognizing who is being tracked, in contrast to other work that uses AI in gait analysis for recognizing individuals[18].

Beyond this difference in purpose, our approach to AI-supported action monitoring also differs methodologically from prior work. Instead of relying on the sensor mat's regular grid geometry both for generating and for interpreting activation patterns, we deliberately decouple the two: a) an aperiodic, direction-unbiased scaffold built from Robinson triangles[11] that diversifies the walking directions in our synthetic training data (Subsection 3.2); and, b) a bank of direction-sensitive[19] convolution kernels[20], instead of the isotropic kernels common in instance segmentation[21]. This combination is designed to exploit directional diversity and to make our neural network sensitive to gait direction rather than only to foot positions (Subsection 3.4); Subsection 4.3 compares it with the isotropic baseline.

The aperiodic scaffold used in Subsection 3.2 builds on existing work: we apply, rather than invent, a much older idea, so it is worth crediting its origin. Figure 1 does so visually. Aristid Lindenmayer introduced recursive string-rewriting systems, now known as L-systems, to model the growth of organisms[3]: a local production rule is applied repeatedly to a seed symbol, independently of any global coordinate frame (Figure 1a). Raphael Robinson used a geometric version of the same rewriting principle in 1971 to prove that a small set of tiles with a matching substitution rule forces any tiling built from them to be aperiodic[4], operating on the golden triangle L and golden gnomon S. Roger Penrose reduced this

construction to its now more widely known two-tile (rhombus) form three years later[5]. These aperiodic tiling principles are not merely mathematical objects, but a relevant option whenever a structured, non-uniform partitioning scheme with measurable practical benefits over a regular grid is required. In modern radio science, e. g., Penrose-tiling-shaped antenna subarrays reduce transmit/receive-module count while preserving wide-angle scan performance[22].

In mechanical engineering, Uwe Grimm et al. computationally characterize the elastic properties of Penrose-rhomb-tiling-based lattice structures and compare them against their periodic counterparts[23]. Figure 1b shows how we benefit from partitioning the plane into tiles with no nontrivial translational symmetry: the two training-data sources introduced in Section 3 reach the same SensFloor hardware grid by different routes.

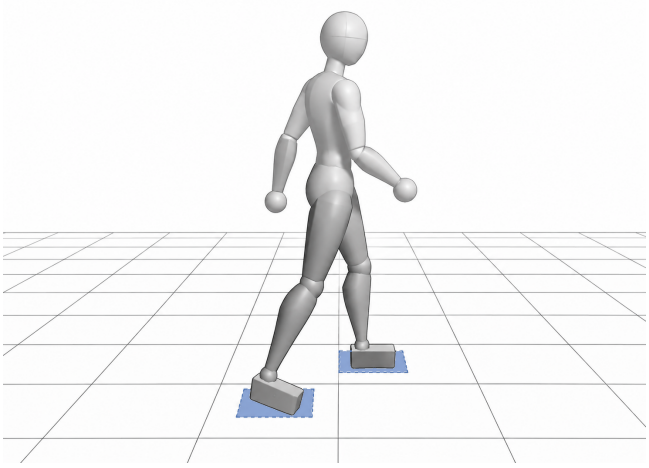


Figure 3. MoCap data are linked to cuboid meshes representing feet. We use the intersections of these meshes with the floor for generating synthetic training data.

Only the simulated data passes through the scaffold, which resamples a single quantity: a walking-direction angle read off one triangle; the labelled real footstep recordings bypass the scaffold entirely and reach the grid unchanged. Unstructured randomness would also erase the 60° orientation bias, but only as a property of a large random sample, whereas the tiling's matching rules guarantee this by construction (Subsection 3.2); they also cluster triangle orientations around the same 36°-multiples that the direction-sensitive kernel bank of Subsection 3.4 is built around. As a result, the walking-direction statistics of the synthetic footstep data inherit the scaffold's direction-unbiased construction, while its positions and physical resolution, like those of the real recordings alongside it, remain exactly those of our sensor hardware.

3 Precise people localization using AI

Our method for detecting the position of people on floor sensor mats is based on the same ideas that form the basis of using AI models for image processing. Instead of images, our AI models process the data matrices of the floor sensor mats. The principle underlying both AI applications, image processing and people detection, is the arrangement of the sensitive areas in a matrix of rectangles. The matrix composed of patches on the floor is comparable to the pixels of a two-dimensional bitmap image. To ensure a good trigger rate, the sensitive rectangular areas are divided alternately along the diagonals and anti-diagonals. The result is a spatial data map wherein each coordinate pair corresponds to two data values: either a measurement at the bottom left and top right or a measurement at the bottom right and top left.

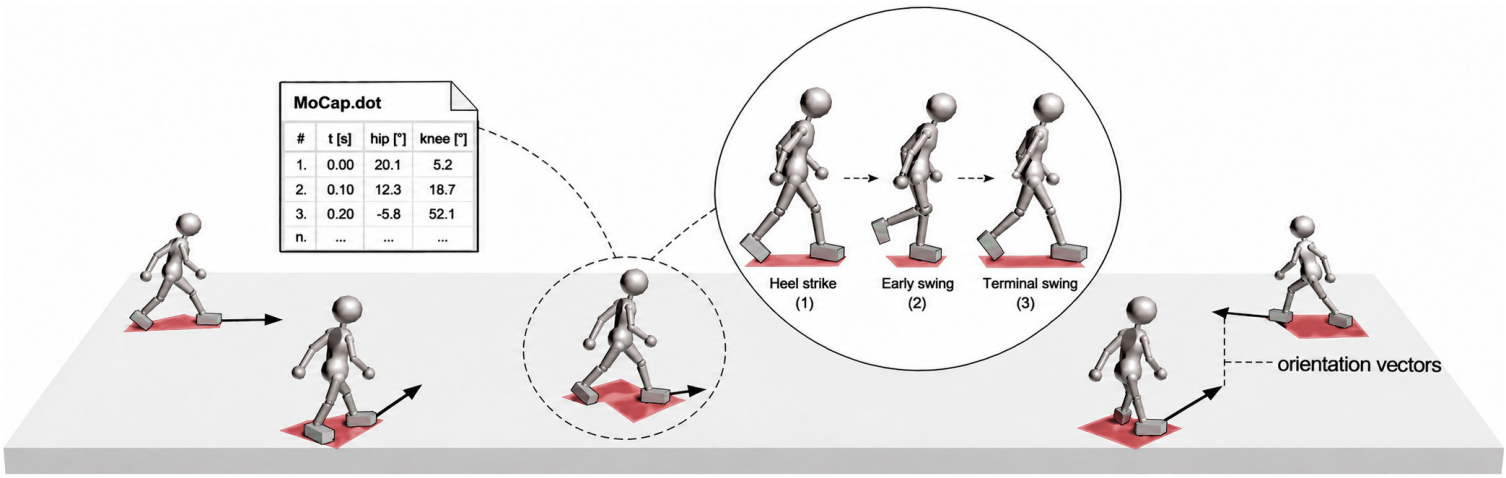


Figure 4. Five simulated humans, with biomechanics modeled according to motion capture files, create gait pattern data.

The processing of data from adjacent patches works segment-by-segment, regardless of the shape of the single patches (or the sensor matrices they contain) which can be triangular (as in our case) or, for example, rhombic.

The only requirement is that all shapes are polygonal and do not overlap, as the neighborhood relationship must be clearly defined. The resulting data are used to train an *instance segmentation network* that can estimate the position of a person's feet walking across the floor sensors by analyzing the pattern generated on the sensor mat from multiple sensors. Using the segmentation masks generated by the network, a weighted average is calculated to estimate the positions of the two feet, labeled F1 and F2 in Figure 2.

Regarding the design of the sensor mats, it should be noted that it is possible and common practice to adapt them to the floor layout without a loss of functionality[24]. In our experiments, we used the floor sensor system "SensFloor" which is primarily used in hospitals for accident detection.

3.1 Realistic Generation of Step Data

Human gait is an inherently *anisotropic* process: a step is a directed movement with a preferred heading. The resulting floor activation pattern, a single foot elongated along its heel-to-toe axis, inherits this directionality rather than being a symmetric or uniformly random pattern. The method we use to enrich data with directionally diverse placements (Subsection 3.2), and our detection network itself (Subsection 3.4) are built on this observation.

In our synthetic data generation procedure, the setting of foot positions is based on a 3D simulation of walking humans triggering signals on simulated SensFloor patches (Figure 3). We modeled the former in *Unity*, due to the flexibility of this software with respect to input and output data[*]. Our patches are described using the so-called *Asset Bundles* in the *Unity Editor*. We export these 3D models to the simulation software *Blender*[†], where we run the movement simulation.

As an alternative to the MoCap files, we also considered using a physics-based simulation, such as the *linear inverted pendulum* (LIP). LIP was originally developed to move biped robots[25] and allows for movement optimizations[26]. Moreover, a LIP simulation produdes a regular movement which, in comparison with the recording of real human movement, helps with identifying abnormalities that indicate pathological conditions[27], which is not relevant to our work. Unlike a clinical study, our model requires only pixel-by-pixel masks as training data, similar to the description of moving objects, their edges, and contours in raster image processing[28].

We link the motion capture data to cuboid meshes, which we use to represent the feet of walking people. The reason for modeling the "brick-shaped" feet, shown in Fig. 3, is that Unity must store a mesh for every object used in Blender for calculating collision events. The patterns left by people walking over the sensors on the floor predominantly depend on the way their feet are placed relative to our sensors. For example, a single foot can activate only the two top triangles of a patch when placed on the upper part. However, it can potentially activate up to eight triangles, which occurs when it is placed in the center of a patch. In the simulation, it is necessary to ensure a good variety of foot placements in relation to the sensors. Therefore, we randomly alter the starting position of the simulated humans every time we spawn another thread to simulate a new human instance in our walking simulation. A snapshot of such a simulation can be seen in Figure 4, where five simulated humans walk across a rectangular grid of floor patches and trigger different activation patterns.

Our algorithm for creating the simulated humans performs the following steps:

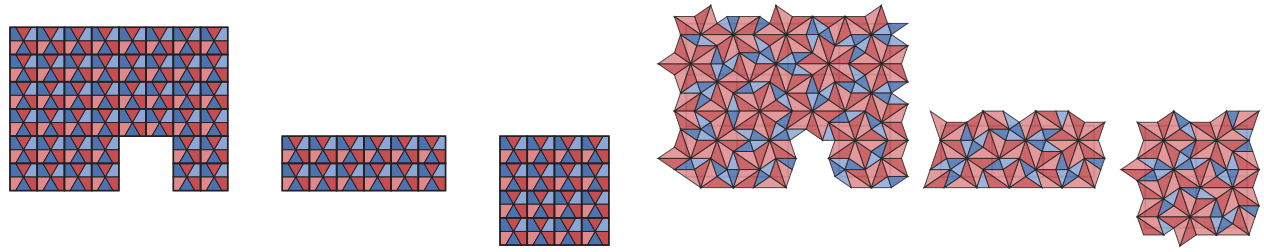
First, the number of simulated humans is chosen. This number varies between 1 and 8. The algorithm then randomly assigns a starting position along any edge of the sensor area to each simulated human. Rather than drawing the walking direction from a fixed angular range, we sample it from an aperiodic Robinson-triangle scaffold, described in Subsection 3.2, specifically to avoid biasing the training data toward a small set of directions. The walking motion for each simulated human is provided through the motion capture file.

3.2 Aperiodic Orientation Sampling

The anisotropy that shapes footstep data also shapes the generative mechanism we use to diversify it. The subdivision rule that produces our Robinson-triangle scaffold belongs to the same family of recursive rewriting systems as an *L-system*[3]: a formal grammar that applies a fixed production rule repeatedly to a seed configuration. In the same way, our decomposition operator repeatedly subdivides a seed configuration of golden triangles. L-systems were introduced to model the growth of organisms and remain the standard formalism for modeling plant development[29]. Plant growth is anisotropic by nature: differences in cell-wall mechanical properties along different directions, such as the orientation of cellulose microfibrils, determine a plant's differential growth

[*]Unity Technologies: Unity (2023), https://unity.com, game development platform

[†]Blender - a 3D modeling and rendering package. Blender Foundation, Stichting Blender Foundation, Amsterdam (2018), https://www.blender.org



(a) Periodic floor layout **(b)** Aperiodic Robinson/Star tiling

Figure 5. The bus sensor-floor footprint (digitized from the as-built module map), partitioned (a) periodically, and (b) with a Robinson/Star-tiling patch clipped to the same footprint.

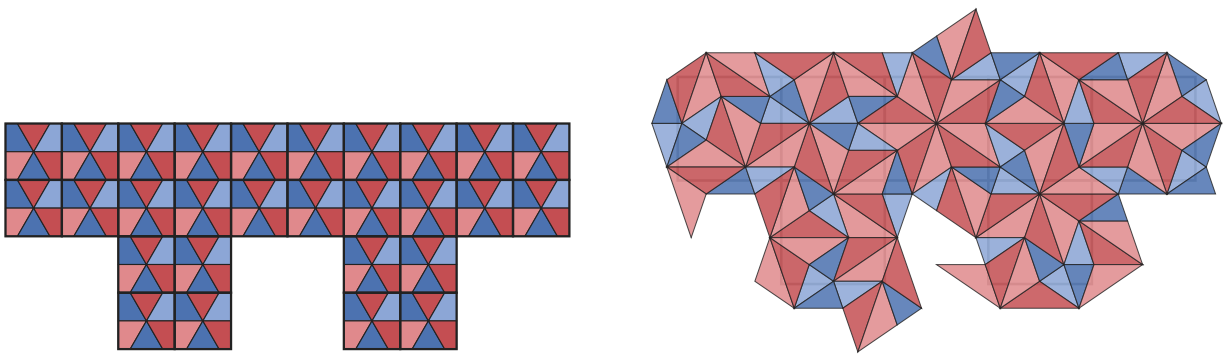
rate and the resulting shape and structural strength[30]. We rely on a substitution grammar with the same generative character since it gives us a directionally rich, non-repeating placement scaffold by construction, rather than one whose direction statistics must be checked for bias after the fact. Since each simulated pedestrian keeps the heading drawn for it throughout its walk, the generated data contains no unrealistic moves such as sequences of complete turnarounds.

The convolutional kernels of our detection network are designed to be sensitive to the direction of a stride (Subsection 3.3), and such a network is only as trustworthy as the directional balance of the data it is trained on. If the training data itself has a structural preference for certain headings, a direction-sensitive kernel risks learning that preference as a property of human gait, rather than recognizing it as an artifact of how the data was generated. To rule out this failure mode by construction, we built an aperiodic tiling scaffold from *Robinson triangles*[11].

A Robinson triangle is one of two golden-ratio isosceles triangles: the *golden triangle* L, with apex angle 36° and two long legs of length $\varphi = \frac{1+\sqrt{5}}{2}$, and the *golden gnomon* S, with apex angle 108° and two short legs of unit length. Their edges carry a matching rule that only allows specific triangles to be joined edge-to-edge. Repeatedly subdividing these triangles and rescaling by φ at each step produces a *Robinson tiling* of the plane that has exactly the properties we need for a realistic choice of directions: no region ever repeats, and neighboring triangles are rotated in multiples of 36°, rather than the $90^\circ/45^\circ$ symmetry of a square grid.

Among the possible tilings that can be built this way, we use the *Star* tiling: starting from a ten-triangle "wheel" seed of L triangles arranged around the origin, we repeatedly apply one subdivision step followed by a $\pi/5$ rotation, and take the union of every second iterate of this process. The Star tiling has *exact* five-fold rotational symmetry: it is invariant under rotation by 72° about its center. Since the symmetry is exact rather than approximate, any sufficiently large patch of the tiling contains triangle orientations with no structural bias toward any particular compass direction.

In our implementation, the axis of symmetry of each L triangle of the ten-triangle wheel seed points in one of the ten directions $k \cdot 36^\circ$, $k = 0, \ldots, 9$ and all ten directions occur equally often. We verified this numerically for up to eight subdivision steps. Sampling a triangle uniformly



(a) Periodic floor layout **(b)** Aperiodic Robinson/Star tiling

Figure 6. An idealized grocery-store footprint (a checkout aisle with two billboard alcoves; unlike the bus, no equally precise as-built map exists for this installation), partitioned (a) periodically and (b) with a clipped Robinson/Star-tiling patch.

from the tiling is therefore equivalent to drawing one of these ten directions with equal probability, which is what our implementation does for each simulated pedestrian. This replaces the heading of our earlier sampler, which added a uniform offset from $[-60°, 60°]$ to a direction pointing away from the pedestrian's entry side[10]; the starting position, the choice of motion-capture file, and all later steps of the pipeline remain unchanged.

To illustrate what this scaffold produces in practice, rather than leaving it as an abstract construction, Figures 5 and 6 show a generated Star-tiling patch clipped to the actual sensor-floor footprints of our two case studies. The shop footprint has no equally precise as-built record, so we use an idealized representative shape instead: a checkout aisle with two alcoves for the billboards. The bus map consists of three disconnected submaps, reflecting gaps in the installation around obstacles such as seating and wheel arches. Next to the periodic tilings of the same floor layouts (left in the figures), it becomes visually obvious that the Robinson tilings never repeat the same local arrangement twice.

Our pipeline involves two different triangular systems that must not be confused with one another. Robinson triangles are golden isosceles triangles with angles that are multiples of $36°$; the SensFloor patches themselves continue to be divided into eight *right* isosceles triangles by their diagonals and anti-diagonals. This sensing geometry is fixed by the hardware and entirely unaffected by our direction choice procedure. The two triangular systems are never overlaid on one another and operate at different stages of the pipeline: the Robinson/Star tiling exists only as an offline, abstract scaffold that we consult once per simulated pedestrian to obtain a single heading angle.

3.3 The Instance Segmentation Network

The goal of our instance segmentation network is to learn which simultaneously active triangles belong to the same physical foot, purely from the spatial structure of the activation pattern rather than any explicit distance rule. To detect and differentiate between the feet of people walking on the floor sensors, we use instance segmentation with *pixel embeddings* and a subsequent *weighted average* calculation to estimate the position.

These implementation decisions are justified for our applications since sensor floor mats are costly and, therefore, often used in small and strategically placed areas. For our supermarket experiments, we focus on the cash register area where two billboards are located. In the public bus, our sensor mats are placed close to the doors. Thus, we are processing dense data that are relevant for localized events such as waiting near a billboard or passenger movement near the bus

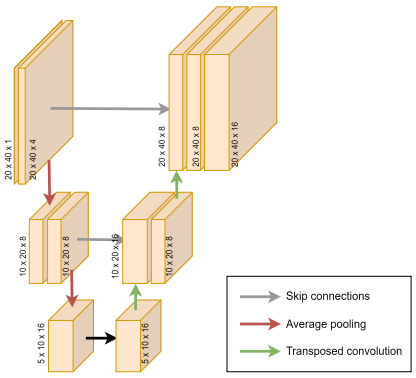


Figure 7. Architecture of our instance segmentation network

doors. The embedding of pixels has shown clear advantages in such scenarios[31] over other AI models, e.g., a Support Vector Machine (SVM) that is suitable when a classification involves a large number of irrelevant features[32].

We embed the individual pixels into an n-dimensional space such that all pixels of one foot instance are near each other and far away from the pixels of other instances. In a subsequent step, the pixel embeddings are clustered to assign them to an instance. We use a *discriminative loss* function and the *mean shift algorithm* for clustering, which have been proven to be useful for training a reliable model without recurrent mechanisms in computer vision[21].

Other instance segmentation networks typically have two stages: 1) the generation of region proposals; and 2) subimage segmentation[33]. In our case, region proposals are not reliable because the activation patterns produced by individual feet consist of a small number of pixels and often occlude each other.

An advantage of the instance segmentation method with pixel embeddings, which we implemented, is that our AI model processes the image as a whole in one single segmentation stage and can better handle complex occlusions. It also does not need bounding boxes to provide additional information to differentiate between instances.

We work with a 10×10 grid of floor patches. Our neural network architecture is shown in Figure 7 and consists of three convolutional layers with a mean pooling between them on the descending branch. For the ascending branch, this structure is mirrored, and the pooling operation is replaced by transposed convolution for upsampling. In addition, two skip connections are used. For the pixel embedding, 16 dimensions are used in the last layer of the network. This network size is sufficient for our purposes. A larger network would be harder to train and more prone to overfitting. The embedding in the 16-dimensional space is done by calculating the loss. Instead of softmax, as commonly used in instance segmentation, we use a discriminative loss function, since

this loss function has shown better performance in detecting objects that are separated into multiple pieces, as required for our floor sensors: it pulls the embeddings towards a cluster center, while dissimilar data are pushed further away from each other. Our implementation of the model is based on U-Net[34] and was trained for 200 epochs on a training dataset with 25918 synthetically generated footstep samples.

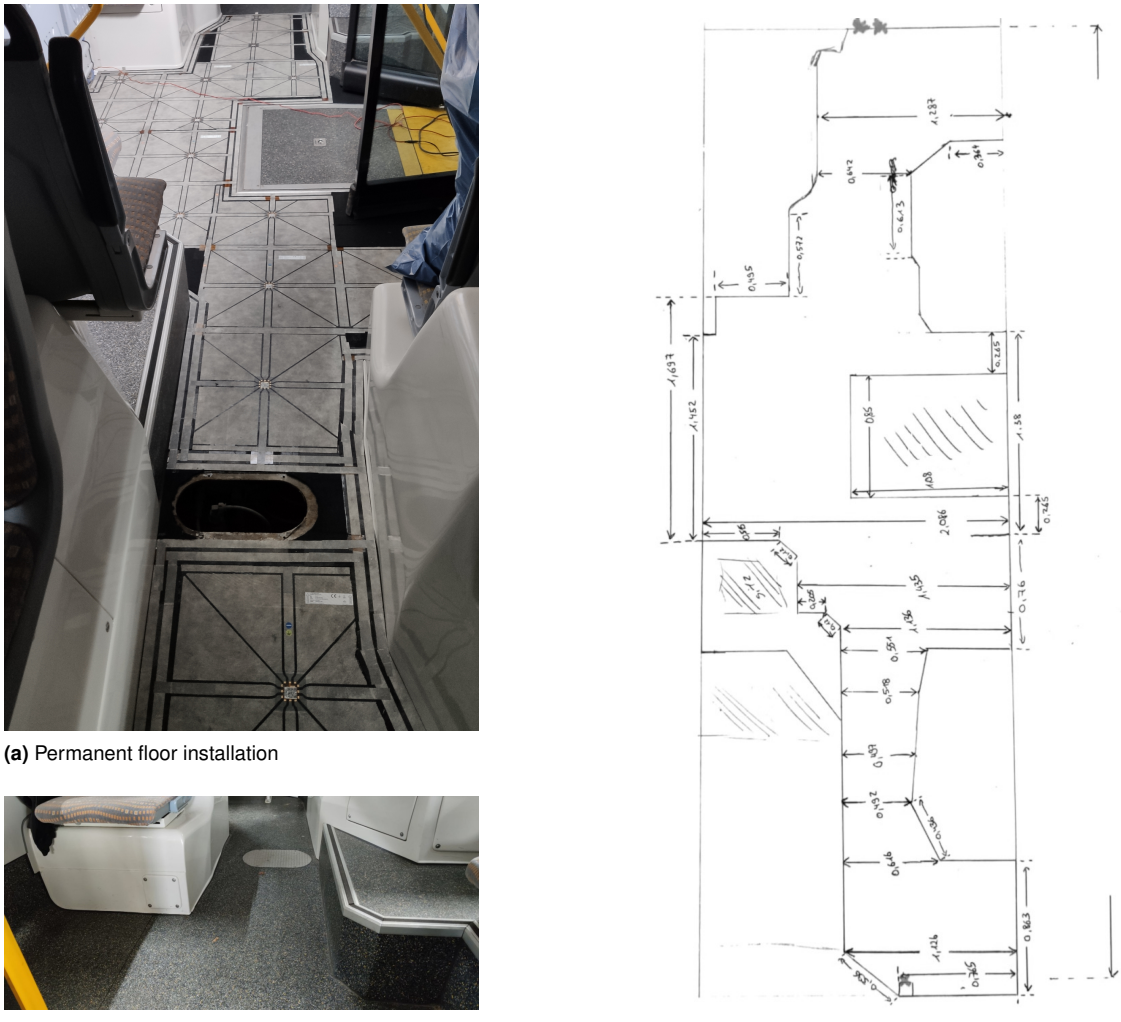
The embeddings are provided by the network and grouped into instances using the mean shift algorithm. The mean shift algorithm starts by selecting any pixel embedding as a cluster center and clusters all other pixel embeddings within a radius around it. In a second step, the mean of all clustered embeddings is calculated and chosen as the new center. All embeddings within a radius around this new center are clustered again. This process is repeated until the position converges and all cluster centers are in a dense region.

After the instance segmentations are obtained, the foot position is calculated using a formula, which originates from the domain of accident detection[35].

$$P = \frac{\sum_{i=1}^{n} w_i p_i}{\sum_{i=1}^{n} w_i} \quad (1)$$

In this formula P is the position of the foot, w_i is the activation capacitance value, and p_i is the position of the centroid of the activated triangle. Calculating the position in this way is accurate since we take into account the different capacitance values. A position will always tend toward activations with higher values, i.e., where the greater part of the foot is placed.

We tested the detection algorithm on a synthetic validation dataset comprising 1541 entries and a hand-annotated dataset with 49 entries. In the synthetic validation dataset, the model scored a precision of 0.937 for an *intersection over union* (IoU) threshold of 0.95, while the precision in the hand-annotated dataset was slightly lower, as shown in Table 1 and Table 2. Given the limited size of the hand-annotated dataset, these real-data results should be interpreted as an initial transferability check. Tables 1 and 2 refer to the pipeline



(a) Permanent floor installation

(c) Covered mats

(b) Layout

Figure 8. Our sensor mats in a city bus.

of our earlier work, with isotropic kernels and the narrow-range heading sampler; Subsection 4.3 compares it with the extended pipeline on a held-out subset of the real recordings.

Table 1. Precision and recall metrics of the network on a synthetic validation dataset.

IoU Threshold	0.500	0.750	0.950
Precision	0.973	0.941	0.937
Recall	0.973	0.941	0.937

Table 2. Precision and recall metrics of the network on a hand-annotated real dataset.

IoU Threshold	0.500	0.750	0.950
Precision	0.989	0.887	0.887
Recall	0.989	0.887	0.887

3.4 Direction-Sensitive Detection via Asymmetric Convolution Kernels

Gait is a directed, non-uniform movement (Subsection 3.1), and our synthetic training data now represents that directionality without a structural bias thanks to an L-system-like substitution scaffold (Subsection 3.2). The results shown in Subsection 3.3 were obtained using isotropic 3×3 convolution kernels throughout [10].

A footstep, however, is not an isotropic pattern: a single foot is elongated along its heel-to-toe axis, and consecutive footfalls of a walking person form a directional sequence across the sensor grid. We therefore replace the isotropic kernel of the first convolutional layer, which operates directly on the full-resolution, un-pooled 20×40 activation grid, and the kernel of the corresponding final layer of the upsampling branch, with a small bank of asymmetric, elongated kernels. This design is inspired by *Asymmetric Convolution Blocks* [19], which strengthen a square kernel's "skeleton" by adding parallel $1 \times k$ and $k \times 1$ branches to it. Unlike ACNet, we do not restrict ourselves to the horizontal and vertical axes: following the same principle used for *Active Rotating Filters* [20], which produce a bank of oriented feature maps by rotating a single learned filter. We rotate one learned 1×3 kernel to a fixed set of five orientations, $\{0°, 36°, 72°, 108°, 144°\}$, which is the same spacing that generates our Robinson/Star training-data scaffold (Subsection 3.2). We then concatenate the five resulting feature maps along the channel dimension before the following pooling operation.

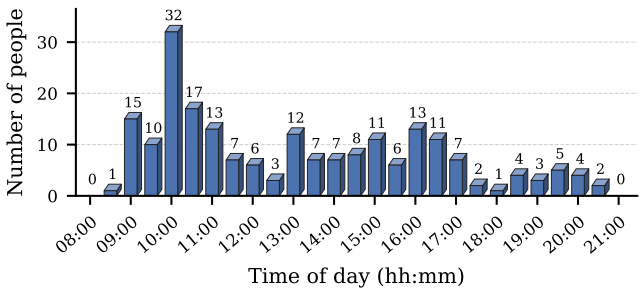
This design choice depends directly on the contribution described in Subsection 3.2. A bank of direction-sensitive kernels is only trustworthy if the walking directions present in the training data are themselves free of a structural bias: training an isotropic kernel on our original $[-60°, 60°]$-orientation angle-limited data was harmless, since an isotropic kernel cannot encode a directional bias. But training a direction-sensitive kernel bank on the same limited data would risk learning that sampling artifact as if it were a movement property. Because the Star tiling's exact five-fold symmetry rules out such a bias by construction (Subsection 3.2), we can introduce direction-sensitive kernels without incurring that risk (see Subsection 4.3 for the empirical comparison). Thus, our contributions introduced in this article are designed to work together, not independently.

Since each of the five oriented kernels has only three taps, the bank has $5 \times 3 = 15$ weights per input channel, compared to 9 for a single isotropic 3×3 kernel over the same input channels. This is a modest increase that remains cheaper than, for example, enlarging the isotropic kernel to 5×5 (25 weights) to capture a comparable spatial extent along the elongated axis.

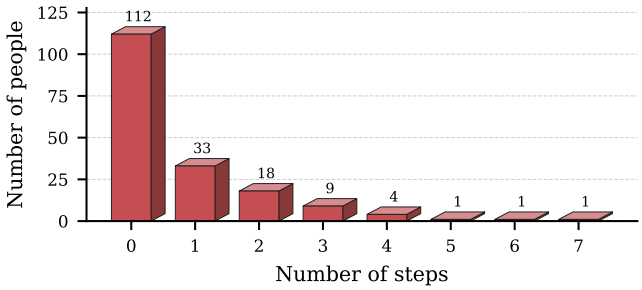
4 Experiments

In addition to synthetic validation (Table 1), we evaluated detection quality using 49 manually annotated real images (Table 2). The result was a precision/recall $= 0.887$ with an IoU of 0.95.

In the following two case studies, we not only count people but also explore movement-related indicators that may be associated with purchasing decisions and other local behavior patterns.



(a) Peak time observation of density on the bus



(b) Step-wise movements while the bus was crowded

Figure 9. Tracking results showing passenger behavior in a public city bus.

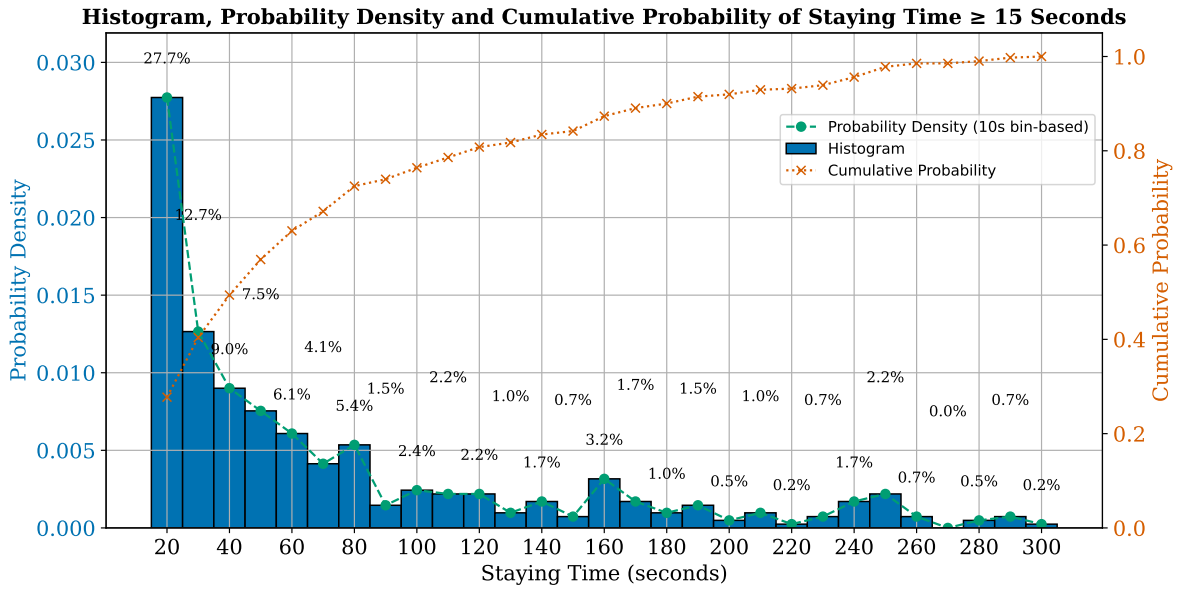


Figure 11. Customer waiting times $\geq 15\,\mathrm{s}$ in 10-second bins, left y-axis: Probability Density, right y-axis: Cumulative Probability.

4.1 Experiments in the City Bus

Figure 8 shows our installation covering the entry/exit area of a city bus, capturing passengers as they board and alight, take a seat, or remain standing in the entry area. Figure 9 shows the tracking results from one month.

We observed peak-time density (Figure 9a) and stepwise movement (Figure 9b), the latter occurring only when fewer than 30 people were on the bus. In this feasibility study, foot movement was classified as a step when two consecutive foot positions of a detected person were more than 30 cm apart, and we used more than 30 passengers as our operational definition of overcrowding. For data protection and operational reasons, we did not record synchronized video during the bus or grocery store trials.

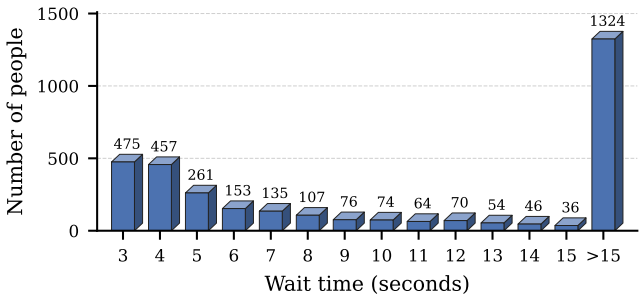
4.2 Grocery Store Experiments

For the second experiment, we installed floor sensors in front of a digital billboard displaying new products at a grocery store and observed the customer behavior for one month. We used heuristic grouping rules: people walking distances shorter than $70\,\mathrm{cm}$ within a given period were treated as showing interest in the advertisement, while those leaving the sensor area within 3 seconds were counted as uninterested customers.

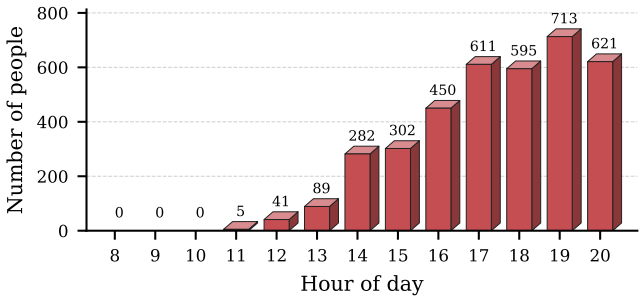
Figure 10a shows a fine-grained classification of durations between 1 and 15 seconds: the number of customers in front of the billboard drops sharply after 5-6 seconds, so only a few customers' attention was held longer. Fewer than 100 customers stayed more than 10 seconds, while over 500 stayed only 2 seconds, which is still longer than a single step takes, so all of them should have noticed the billboard.

A statistical analysis of standing times is shown in Figure 11, focused on people detected in front of the billboard for more than $15\,\mathrm{s}$, which we interpret as having viewed the ad (some false positives remain, e. g., people standing nearby). This is supported by the approximately periodic length of stay: grouped into $10\,\mathrm{s}$ bins, many people turned away around $80\,\mathrm{s}$, $160\,\mathrm{s}$, and $240\,\mathrm{s}$, which matches the repetition interval of the looping advertisement and suggests a plausible mapping between foot movements and customer behavior.

The average customer count over the month, shown in Figure 10b and verified against our on-site transcript, peaked



(a) People grouped at the billboard by wait time



(b) Hourly count of customers

Figure 10. Shop visitor distribution during business hours.

Table 3. Comparison of the combined direction-unbiased data enrichment (Subsection 3.2) and direction-sensitive kernels (Subsection 3.4) with the baselines: precision and recall at an IoU threshold of 0.5 on 16 held-out real recordings, one training run per condition.

#	Enrichment / heading sampler	Kernel	Tests	Prec.	Rec.
1	scarce real data only, no enrichment	isotropic	absolute floor	0.875	0.875
2	narrow range $[-60°, 60°]$ [10]	isotropic	prior published method	1.000	1.000
3	naive uniform $360°$	isotropic	"widening the range alone" claim	0.969	0.969
4	Robinson/Star scaffold	anisotropic	full combined method	0.969	0.969

in the evening after office hours. Duration measurements from tracking customers' feet are a further outcome beyond what simple cash-register counts would give: staying duration, customer count, and inter-foot distance (crowdedness) together make our data suitable for *partial information decomposition*[36], which classifies such values as unique, redundant, or synergistic.

4.3 Effects of the Improvements

The two methodological contributions described in Subsections 3.2 and 3.4 are designed to work together, not independently: a bank of direction-sensitive kernels is only trustworthy if it is trained on direction-unbiased data (Subsection 3.4). We therefore do not evaluate either contribution on its own, but compare the full combined method with three baselines (Table 3): a scarce-real-data baseline (1), the original narrow-range heading sampler this work extends [10] (2), and a naive full-$360°$-uniform heading sampler as a control for whether merely widening the sampling range is sufficient (3). Row 1 is trained on 33 of the 49 hand-annotated real recordings; rows 2–4 are each trained on 25918 synthetic examples from the same simulation pipeline, which differ only in the heading sampler and, for row 4, in the kernels. All four models were trained for 200 epochs with identical hyperparameters and are evaluated on the remaining 16 real recordings, which none of them saw during training.

Data generation and training ran on two machines. The synthetic training data for conditions 3 and 4 were generated with the Unity simulation on a laptop with an Intel Core Ultra 7 processor (8 cores, up to $5.0\,\mathrm{GHz}$) and $32\,\mathrm{GB}$ of RAM. Since the simulation runs in real time, generating the 25918 examples of one condition took about three hours; both conditions were generated in parallel. All models were trained on a workstation with an AMD Ryzen Threadripper PRO 7965WX processor (24 cores, 48 threads, up to $5.3\,\mathrm{GHz}$, $128\,\mathrm{MB}$ L3 cache), $64\,\mathrm{GB}$ of RAM, and two NVIDIA RTX 4000 Ada Generation GPUs with $20\,\mathrm{GB}$ of memory each, running Fedora Linux, Python 3.14, and PyTorch 2.14 with CUDA 13. Training for 200 epochs took about 52 minutes for condition 2 and about 58 minutes each for conditions 3 and 4, which were trained in parallel on the two GPUs; condition 1, with only 33 training examples, took less than two minutes. Repeating conditions 1–4 with three random seeds each, with three runs sharing one GPU, took about 2 h 40 min per run for conditions 2–4 and a few minutes per run for condition 1. Our experiments can be reproduced and transferred to other installations using the publicly available code and data[‡].

The results show that our synthetic data enrichment pays off: all models trained on synthetic data outperform the real-data-only baseline, with a precision and recall between 0.97 and 1.00 instead of 0.88 at an IoU threshold of 0.5, and 0.81–0.91 instead of 0.69 at 0.75. Among the synthetic conditions, however, we observe no consistent ranking: the narrow-range sampler (2) scores highest at an IoU of 0.5, the naive uniform sampler (3) at 0.75, and the full combined method (4) did not outperform the prior method in the experiment. With 16 test recordings, each of which accounts for $1/16$ of the score, and a single training run per condition, these differences are within the noise of this evaluation. We see two plausible reasons why direction balance is not visible in the table. First, the real test recordings stem from two deployments in which the geometry favors a few walking directions. So, a test set of this kind can hardly reward direction balance in the training data. Second, the heading sampler also changes the data beyond the directions: with headings independent of the entry side, more simulated pedestrians leave the sensor area early, so the training examples of conditions 3 and 4 contain fewer simultaneous feet (2.2 on average) than those of condition 2 (2.9). The final proof that direction-unbiased enrichment combined with direction-sensitive kernels leads to the best possible footstep detection therefore requires a larger, orientation-complete test set from a larger floor mat and repeated training runs.

5 Conclusion

One research question studied in this work was how computers can analyze local movement patterns in public spaces when camera recordings are not available. Using floor

[‡] https://github.com/JanDuennweber/walking-in-every-direction

sensor mats on a bus and in a supermarket, and training with biomechanically simulated data, we created an AI model for privacy-aware gait analysis. Our results suggest that ML can detect footsteps on floor sensor mats with promising precision, that synthetic biomechanics-based data are useful for training, and that sensor placement in small, strategically chosen areas strongly affects the usefulness of the measurements. We further address the risk of a directional bias in synthetic training data by sampling simulated walking directions from an aperiodic Robinson-triangle scaffold rather than a fixed angular range, and we introduce a matching bank of direction-sensitive convolution kernels designed to exploit this directional diversity for gait classification, rather than treating footsteps as an isotropic pattern. Compared with training on scarce real data alone, synthetic enrichment improved footstep detection clearly; the combination of direction-unbiased enrichment and direction-sensitive kernels, however, performed at the same level as our earlier narrow-range sampler, but was not better on the small real test set available to us. While floor sensor mats are not suitable for fast large-area tracking, they offer a privacy-preserving option for localized planning support. Given the small annotated real dataset and the heuristic nature of the case studies without synchronized video ground truth, the behavioral findings should be understood as indicative; future work should evaluate this combination on a larger, orientation-complete real test set with repeated training runs, extend the real-world validation, and compare our floor patches to alternative privacy-aware people tracking approaches.

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